



CHIJ ST. THERESA'S CONVENT
PRELIMINARY EXAMINATION 2026
SECONDARY 4 EXPRESS / 5 NORMAL (ACADEMIC)

CANDIDATE
NAME

CLASS

INDEX
NUMBER

ADDITIONAL MATHEMATICS

4049/2

Paper 1 (Part A)

21 August 2026

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your index number, and name on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 90.

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!} = \frac{n(n-1)\dots(n-r+1)}{r!}$.

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2}ab \sin C$$

- 1 A solid cuboid has a square base of side $(3 + 2\sqrt{3})$ cm.

The volume of the cuboid is $(141 + 81\sqrt{3})$ cm³.

Without using a calculator, express the total surface area of the cuboid in the form $(a + b\sqrt{3})$ cm², where a and b are constants. [7]

- 2 (a) Find $\frac{d}{dx}\sqrt{2x^2-3x+1}$, leaving your answer in the form $\frac{ax+b}{c\sqrt{2x^2-3x+1}}$, where a , b and c are integers to be found. [2]

(b) Hence find $\int \frac{24x-18}{\sqrt{2x^2-3x+1}} dx$ [2]

- 3 The acute angles A and B are such that $\tan(A+B) = 6$ and $\tan B = 2$. Without using a calculator, find the exact value of $\cos A$. [5]

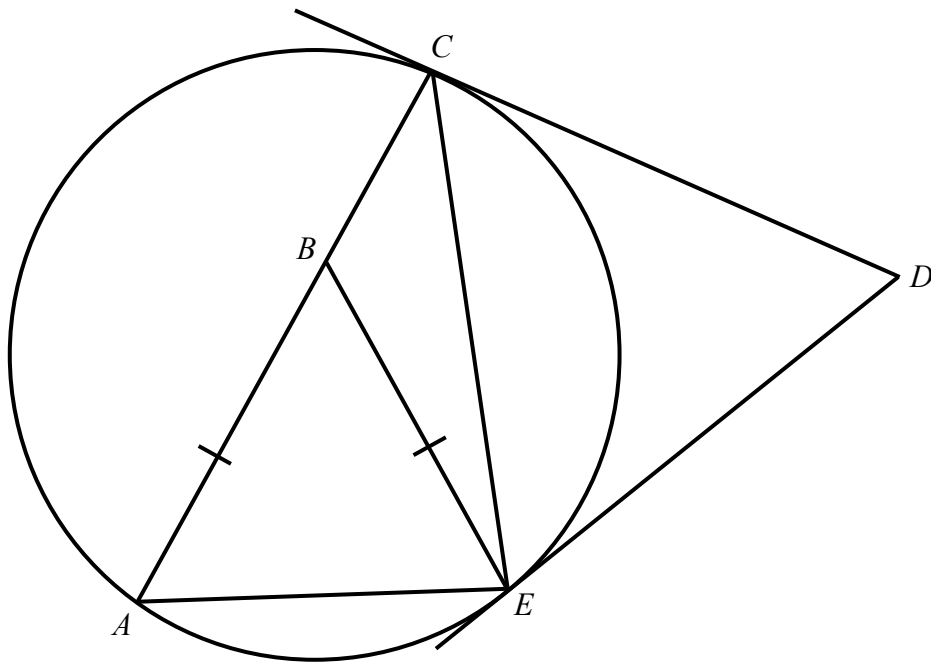
- 4 A circle has a diameter AB . The point A has coordinates $(1, -2)$ and the equation of the tangent to the circle at B is $y = x - 15$.

(i) Find the coordinates of B . [4]

(ii) Find the radius and the coordinates of the centre of the circle. [3]

- 5 (i) Given that the coefficient of x^2 in the expansion of $(2-x)^2(1-px)^8$ is 81, find the two possible values of the constant p . [5]

- (ii) For each value of p , find the coefficient of x^3 in the expansion $(1-px)^8$. [3]



The diagram shows a circle passing through the vertices of a triangle ACE . The tangents to the circle at C and E intersect at the point D . The point B lies on AC such that $AB = BE$.

- (i) Prove that B, C, D and E lie on a circle.

[5]

(ii) Hence, show that BD is an angle bisector of angle CBE .

[2]

- 7 [For a sphere of radius r units, volume = $\frac{4}{3}\pi r^3$ and surface area = $4\pi r^2$.]

A spherical balloon is being inflated with helium gas at a constant rate of 200 cm^3 per second. The balloon is initially empty.

- (i) Find the radius of the balloon after one minute. [2]

- (ii) Find the rate at which the surface area of the balloon is increasing after one minute. [4]

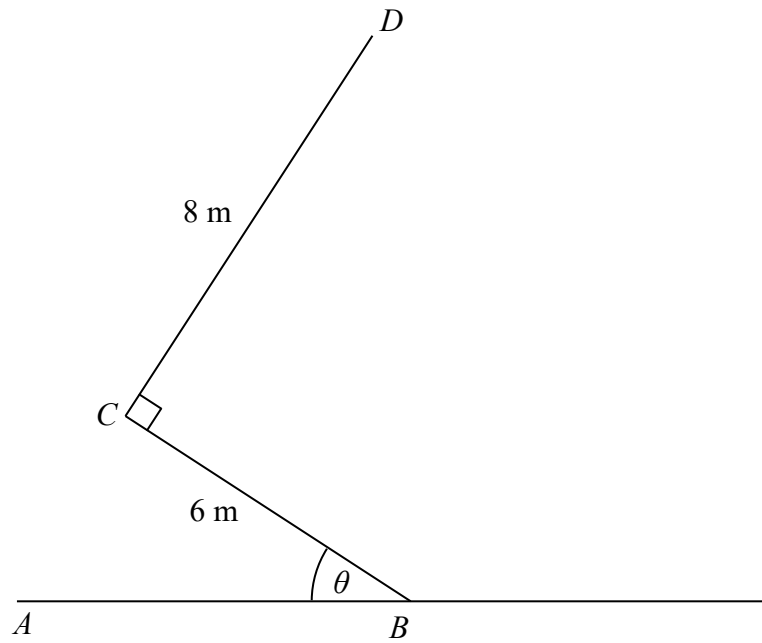
- 8** An open cylindrical container has a base radius of r cm and a height of h cm.
The total surface area of the container is $320\pi \text{ cm}^2$.

(i) Show that the volume of the cylindrical container, $V \text{ cm}^3$, is given by $V = 160\pi r - \frac{1}{2}\pi r^3$. [3]

Given that r can vary,

(ii) find the stationary value of V and determine its nature. [4]

- 9 (a) The equation of a curve is $y = px^2 + qx + r$, where p , q and r are constants. The curve intersects the y -axis at $(0, 7)$ and has a minimum point at $(2, -1)$. Find the values of p , q and r . [5]
- (b) Find the set of values of the constant k for which the curve $y = kx^2 + 6x + k - 1$ lies completely above the line $y = -9$. [5]



The diagram shows two rods BC and CD rigidly joined at C so that angle $BCD = 90^\circ$. The lengths of BC and CD are 6 m and 8 m respectively and angle $ABC = \theta^\circ$.

- (i) Show that the shortest distance of D to AB , L , is given by

$$L = 8 \cos \theta + 6 \sin \theta.$$

[2]

- (ii) By expressing L in the form $R \cos(\theta - \alpha)$ where $R > 0$ and $0 < \alpha < 90^\circ$, find the maximum possible length of L and the corresponding value of θ . [6]

- (iii) Find the value of θ for which $L = 7$ m. [2]

11 Given that $y = p \cos 3x + q$, where p and q are positive integers,

(i) state the period of y . [1]

Given that the maximum and minimum values of y are 7 and -3 respectively, find

(ii) the amplitude of y , [1]

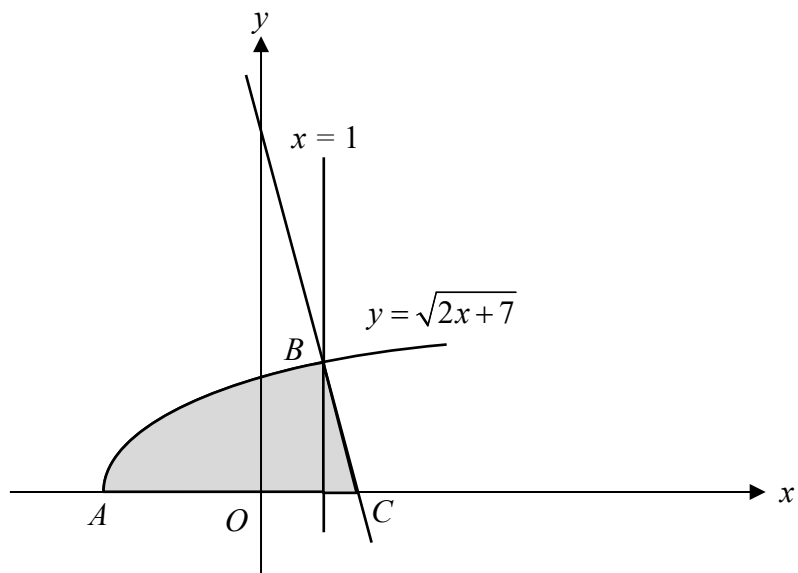
(iii) the value of p and of q . [2]

Using the values of p and q found in part (iii),

(iv) find, in degrees, the smallest positive value of x for which $y = 0$. [2]

- (v) sketch the graph of y for $0^\circ \leq x \leq 180^\circ$.

[3]



The diagram shows part of the curve $y = \sqrt{2x+7}$, meeting the x -axis at the point A and the line $x = 1$ at the point B . The normal to the curve at B meets the x -axis at the point C . Find

- (i) the coordinates of C ,

[4]

(ii) the area of the shaded region.

[6]

~~~ End of Paper ~~~